

ECE 268 Project Progress Report

Lightweight Block Ciphers for Constrained Devices

Narain Shriraam M S (A69041741)

Karthik N R

Tushar Nasery

June 3, 2026

Project Summary

Goal: Implement SIMON-64/128 and GIFT-128/128 with encryption, decryption, key schedules, and CBC/CTR modes; validate against published test vectors; benchmark against an AES-128 baseline (code size, cycles/byte, key-schedule cost) for IoT/RFID settings where AES is too costly.

Status: All implementations are complete and correct. 16/16 test cases pass including spec-mandated ECB vectors. Benchmarks and code-size measurements are collected.

1 Work Completed

Implementations

- **Repository layout:** `simon64.c/h`, `gift128.c/h`, `modes.c/h`, `test_vectors.c`, `bench.c`, and `Makefile`.
- **SIMON 64/128:** 64-bit block, 128-bit key, 44 rounds (ePrint 2013/404). Uses z_3 key schedule ($m=4$), big-endian block I/O. Feistel round function uses circular shifts (1, 2, 8) and bitwise AND/XOR.
- **GIFT-128/128:** 128-bit block/key, 40 rounds (ePrint 2017/622). Nibble-based SPN with 4-bit S-box, 128-bit bit-permutation, 6-bit round constants, and key update. Precomputed round keys `rk[40]` [2].
- **Modes:** Cipher-agnostic CBC (PKCS#7 padding) and CTR (big-endian counter) via function-pointer interface `block_fn_t`. Any cipher plugs in without modification.
- **Test harness:** 16 test cases covering SIMON ECB, GIFT ECB (2 spec vectors), CBC round-trips for both ciphers, CTR round-trips, and PKCS#7 padding edge cases (1-byte, block-aligned).
- **Benchmark:** `bench.c` measures key-schedule cost (ns), ECB and CTR throughput (cycles/byte) for SIMON, GIFT, and AES-128 (CommonCrypto hw-accelerated baseline).

Test Vector Validation

Cipher	Block / Key	Spec vectors	CBC/CTR	PKCS#7
SIMON 64/128	64 / 128	Pass	Pass	Pass
GIFT-128/128	128 / 128	Pass (2/2)	Pass	—

Benchmark Results (Apple Silicon, -O2, 64 KiB)

Cipher	Key Sched (ns)	ECB (cpb)	CTR (cpb)	.text (B)
SIMON 64/128	38	10.9	10.8	356
GIFT-128/128	488	358.8	338.9	4476
AES-128 (hw)	448	1.8	1.6	—

Analysis: SIMON’s Feistel structure with simple bitwise operations yields a tiny code footprint (356 B) and competitive throughput (≈ 11 cpb). GIFT’s nibble-based SPN permutation is significantly slower in software

(≈ 340 cpb) but is designed for minimal hardware gate count (~ 1300 GE per literature). AES-128 with hardware AES-NI achieves <2 cpb but is unavailable on the target constrained platforms.

2 Challenges and Issues Encountered

- **SIMON variant confusion:** The original implementation used SIMON 64/64 ($m=2$, 32 rounds, z_0), a non-standard variant absent from the spec appendix. Corrected to SIMON 64/128 ($m=4$, 44 rounds, z_3) which has a published test vector.
- **GIFT nibble packing:** The byte-to-nibble conversion had swapped high/low nibbles, causing all encrypt outputs to mismatch. Fixed by aligning with the reference C++ implementation's ordering.
- **Key schedule constants:** Ensuring the correct z -sequence constant (`0xfc2ce51207a635db`) for z_3 required cross-checking against Crypto++ and refinery implementations.

3 Plans for Final Submission

Remaining work:

1. Final report: expand analysis with gate-count references, Feistel vs. SPN trade-off discussion.
2. Presentation: 10–12 slides with architecture diagrams and benchmark bar charts.
3. Stretch: compare at additional data sizes (1 KB, 16 KB) and add a third cipher (SPECK-64 or PRESENT-80).

Member	Tasks
Narain Shriraam M S	GIFT-128 implementation, benchmark scripts, final report
Karthik N R	SIMON-64 implementation, AES baseline, results plots
Tushar Nasery	Test harness, Makefile, modes, slides

Feasibility and Direction

The project is on track. All core deliverables (two cipher implementations, CBC/CTR wrappers, test vector validation, and AES-baseline benchmark) are complete and passing. The remaining work is report writing and presentation preparation. The benchmark data confirms the design rationale: SIMON offers the best software performance among lightweight ciphers, while GIFT targets hardware efficiency.

References: Bogdanov et al., PRESENT (CHES 2007); Beaulieu et al., SIMON/SPECK (ePrint 2013/404); Banik et al., GIFT (ePrint 2017/622).